

Understanding Fourier Transformation

Student ID

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1 Introduction

The Fourier transform is a mathematical tool used to analyze functions and signals in the frequency domain. It allows us to represent a function as a sum of sinusoidal components. The Fourier transform of a function $f(t)$ is denoted as $\mathcal{F}\{f(t)\}$ or simply $F(\omega)$.

2 Continuous Fourier Transform

The continuous Fourier transform is defined as:

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt \quad (1)$$

In [Equation 1](#), $F(\omega)$ is the complex Fourier transform of the function $f(t)$, ω is the angular frequency in radians per second, $f(t)$ is the time-domain function, and i is the imaginary unit.

3 Inverse Continuous Fourier Transform

The inverse continuous Fourier transform allows us to reconstruct the original function $f(t)$ from its Fourier transform $F(\omega)$ as shown in [Equation 2](#).

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)e^{i\omega t} d\omega \quad (2)$$

4 Properties of the Fourier Transform

The Fourier transform has several important properties as shown in [Equation 3 - 6](#).

$$\mathcal{F}\{af(t) + bg(t)\} = aF(\omega) + bG(\omega) \quad (\text{Linearity}) \quad (3)$$

$$\mathcal{F}\{f(t - t_0)\} = F(\omega)e^{-i\omega t_0} \quad (\text{Time-shifting}) \quad (4)$$

$$\mathcal{F}\{e^{i\omega_0 t} f(t)\} = F(\omega - \omega_0) \quad (\text{Frequency-shifting}) \quad (5)$$

$$\mathcal{F}\{f(at)\} = \frac{1}{|a|} F\left(\frac{\omega}{a}\right) \quad (\text{Scaling}) \quad (6)$$

5 Discrete Fourier Transform (DFT)

In practice, we often work with discrete signals. The Discrete Fourier Transform (DFT) is used to analyze such signals. The DFT of a discrete signal $x[n]$ is defined as in [Equation 7](#).

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-i2\pi kn/N} \quad (7)$$

Here: $X[k]$ is the DFT of the signal, $x[n]$ is the discrete signal, N is the number of samples in the signal, and k is the frequency index.

6 Conclusion

The Fourier transform is a powerful tool for analyzing signals in the frequency domain. It allows us to decompose complex signals into simpler sinusoidal components, making it useful in a wide range of applications, from signal processing to image analysis.

Tasks

1. Replicate at least 1 inline mathematical expressions
2. Replicate [section 4](#) and [section 5](#) with
 - the mathematical expressions and
 - the references to the expressions